ProMaint

Predictive Maintenance and Failure Prediction System

A project for predicting machine failures and how to maximize usage of machines in factories.

Project Analysis

Problem:

Machines operate 24/7 and hardly stop working, especially in large factories. However, continuous use can lead to breakdowns that disrupt production, and result in wasted time and money. Additionally, contacting a maintenance manager after a breakdown can be time-consuming and may require additional equipment, causing further exacerbating production delays. This highlights the need for our project idea.

Goal:

predict when machines will fail and suggest predictive maintenance

Scope:

The system is designed to be applied within the following sectors:

- **Large Factories/Manufacturing Plants:** Focusing on high-value production machinery where unexpected downtime is costly.
- **Power Firms/Energy Sector:** Targeting critical infrastructure equipment.
- **Hospitals/Healthcare Institutions:** Applied to essential, life-critical equipment.

Limitations:

The project will not encompass:

- **Failure type diagnosis**: The project will not identify the type of failure, and therefore, it will not offer solutions for addressing issues in the machinery.

Tools used:

Python for data cleaning and exploratory data analysis (EDA).

SQL for building the database.

Machine learning models: a variety of models including **Random Forest and XGBoost**, as well as advanced deep learning models such as **LSTM and Transformers.**

Evaluation metrics: for testing model performance like precision, recall and f1 score.

Deployment tools: Streamlit- Flask, FastAPI-GitHub for monitoring updates on the project.

Project timeline:

1. Exploratory Data Analysis (EDA) & Data Understanding (4 days):

- **Objective:** To understand the nature of the data, identify general patterns, discover issues, and examine the relationships between variables.

2. Data Cleaning & Preprocessing (1 week):

- **Objective:** To prepare the raw data, making it suitable and clean for subsequent analysis and modeling.

3. Feature Engineering (3 days):

- **Objective:** To extract intelligent indicators (features) from the raw data that will help the model make accurate predictions to get a ready-to-train dataset.

4. Model Training & Modeling (1 week):

- **Objective:** To train a prediction model for machine failure or Remaining Useful Life (RUL).

Steps:

- Split the data into train, validation, and test sets.
- Start by experimenting with different models (e.g., Random Forest, XGBoost).
- Test advanced time-series models (e.g., LSTM, GRU, Transformer).
- Perform **Hyperparameter Tuning** to optimize model performance.
- Evaluate performance using appropriate metrics:
  - For Classification (Failure/No Failure): **Precision, Recall, F1-score.**

- For RUL (Regression): **RMSE, MAE.**

5. Deployment (1 week):

- **Objective:** To make the model fully operational and functional on real-time data.

Steps:

- Build an **API interface** (using tools like Flask / FastAPI) to receive new machine data.
- Configure the system to perform predictions in real-time.
- Upon predicting an upcoming issue, the system should: send an **alert** or a recommendation for **preventive maintenance**.

Deliverables (Outputs):

A readily accessible Predictive Maintenance API for predicting machine failure and suggesting maintenance scheduling tasks.

Stakeholder Analysis:

End Users:

Stakeholder	influence	Reason
Maintenance team	high	Need for accurate alerts to schedule maintenance tasks (lessen time and effort)
Operation managers	high	Need for sustainable production using accurate results from the system
Machine operates	medium	Need for simple user interface to understand the system and when to address issues for maintenance team

Decision Makers:

Stakeholder	influence	Reason

CFO	Medium	Need for driving growth, with saving time and costs for maintenance as well as gaining income which will be achieved through the system
CTO	Medium	Handles system infrastructure and how to integrate it with the factors' systems

Project implementation team:

Stakeholder	influence	Reason
Data science team	high	To get product with high accuracy and better results

Database Design

Introduction

The database is designed to store data collected from sensors attached to industrial machines. This data includes information such as temperature, torque, and tool consumption during operation.

It is used to analyze the condition of the machines and to train a model that predicts when failures are likely to occur, helping improve maintenance efficiency and reduce unexpected downtime.

Tables Overview

The database consists of a single main table that contains all the readings from the machines:

Table name: measurements

This table includes all the values recorded by the sensors, along with product information and the type of failure, if any.

Table Details

Column Name	Data Type	Description
UDI	INTEGER	A unique identifier for each machine (Unique Device ID)
Product ID	VARCHAR	The product identification number being manufactured
Type	CHAR(1)	The type of process or product category, possible values are: L (Low), M (Medium), H (High)
Air Temperature	FLOAT	Air temperature during operation
Process Temperature	FLOAT	Process temperature inside the machine
Rotational speed [rpm]	INTEGER	Machine rotation speed in revolutions per minute (Revolutions Per Minute - RPM)
Torque	FLOAT	Torque value
Tool wear	INTEGER	Tool wear during use
Machine Failure	BOOLEAN	Indicates whether a machine failure occurred (1 = Yes, 0 = No)
TWF	BOOLEAN	Tool Wear Failure – Failure due to tool wear
HDF	BOOLEAN	Heat Dissipation Failure – Failure due to overheating

PWF	BOOLEAN	Power Failure – Failure due to a power malfunction
OSF	BOOLEAN	Overstrain Failure – Failure due to overloading
RNF	BOOLEAN	Random Failure – Failure of an unspecified or random cause

Relationships

The current design is based on a single table (measurements), so there are no internal table relationships in the basic version of the document.

ER Diagram

MEASUREMENTS		
BIGSERIAL	measurement_id	PK
INT	udi	
VARCHAR	product_id	
CHAR	type	
DOUBLE	air_temperature	
DOUBLE	process_temperature	
INT	rotational_speed	
DOUBLE	torque	
INT	tool_wear	
BOOLEAN	machine_failure	
BOOLEAN	twf	
BOOLEAN	hdf	
BOOLEAN	pwf	
BOOLEAN	osf	
BOOLEAN	rnf	

UI/UX Design

Overview

The system is designed to monitor the performance of machines on production lines through an interactive interface that displays sensor readings (such as temperature, speed, torque, and tool wear) along with predictions of potential future failures.
The purpose of the design is to help maintenance engineers make fast, data-driven decisions and reduce unplanned downtime.

Main Interfaces

Dashboard:

- Displays the status of all machines in real-time using color codes (Green – Yellow – Red).
- Includes a summary of readings such as temperature, torque, speed, and tool wear.

- Its purpose is to provide an overview of the machines' conditions in the factory.

Machine Details:

- Contains detailed data for a specific machine, such as temperature, speed, torque, and tool wear.
- Includes charts showing how the readings change over time.
- Its purpose is to analyze the performance of a particular machine and monitor its condition accurately.

Failure Predictions:

- Shows predictions generated by the model, such as the probability of machine failure and the expected time until a failure occurs.
- Its purpose is to help the maintenance team make timely preventive maintenance decisions.

Alerts & Maintenance:

- Displays alerts when readings exceed normal values, such as high temperature or increased tool wear.
- May also include suggestions for the next maintenance schedule.
- Its purpose is to enable the team to respond quickly to avoid failures and minimize losses.

Design Elements

Colors Used:

- **Green:** Machine is operating efficiently.
- **Yellow:** Requires monitoring or upcoming maintenance. • **Red:** Failure or expected fault.

Visual Representation:

- **Line Chart:** To show changes in temperature or speed over time.
- **Bar Chart:** To compare torque with tool wear.
- **Gauge Chart:** To display the machine's status as a performance percentage (e.g., 80%).
- **Cards in the Dashboard:** Summarize the number of operating or stopped machines.